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BTECH PROJECT

A CLOUD BASED DOCUMENT MANAGEMENT SYSTEM

*A Bachelor of Technology project submitted to the department of Technology
SEAS-IUC in partial fulfillment of the requirement for the award of a Bachelor
of Technology (BTECH) degree in Cloud Computing.*

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CERTIFICATION

This is to certify that this report was written by **AWARA DIANA TENGWI** and accepted as partial fulfilment of the requirements for the award of a Bachelor of Technology (B. Tech) degree in Computer Engineering/Cloud Computing.

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DEDICATION



**TO MY
FAMILY**

ACKNOWLEDGMENTS

We have taken efforts in this project. However, it would not have been possible without the kind support and help of many individuals and organizations. We would like to extend our sincere thanks to all of them. The accomplishment of this work required the assistance of some people. Failing to quote them individually here, they receive our most sincere thanks. We are particularly grateful to:

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ABSTRACT

This report presents DocuTrack Pro, a cloud-based system designed to improve how organizations handle documents. It solved common problems like slow manual workflows, lack of tracking, and limited teamwork. The system was built using modern tools like React, TypeScript, and Tailwind CSS, and it runs on Amazon Web Services (AWS), using services such as Lambda for the backend coding, API Gateway for routing the API functions, S3 for static hosting, and RDS for database storing. With a serverless design, it is easy to scale, maintain, and cost-effective. The system supports three user roles—Clients who upload documents, Processing Staff who work on them, and Deputy Directors who manage and review. Key features include automatic document routing, real-time updates, secure access control, alerts, and reporting. It was developed using Agile methods with feedback at each step and deployed using AWS's free tier to keep costs low. Tests showed it improved processing speed, transparency, and user experience. DocuTrack Pro offers a modern, secure, and flexible solution for managing documents in any organization.

Keywords: Cloud Computing, Document Management System, AWS Services, Serverless Architecture, Real-Time Tracking .

RESUME

Ce rapport présente DocuTrack Pro, un système basé sur le cloud conçu pour améliorer la gestion des documents au sein des organisations. Il répond à des problèmes courants tels que la lenteur des processus manuels, l'absence de suivi efficace et le manque de collaboration entre les équipes. Le système a été développé à l'aide d'outils modernes comme React, TypeScript et Tailwind CSS, et fonctionne sur l'infrastructure Amazon Web Services (AWS), en utilisant des services comme Lambda pour le développement du backend, API Gateway pour le routage des fonctions API, S3 pour l'hébergement statique, et RDS pour le stockage des données. Grâce à une architecture serverless, il offre une solution évolutive, facile à maintenir et rentable. Le système prend en charge trois rôles d'utilisateurs distincts : les clients qui téléversent les documents, le personnel chargé du traitement qui travaille sur les documents, et les directeurs adjoints qui supervisent et valident les opérations. Parmi ses fonctionnalités principales, on retrouve le routage automatique des documents, les mises à jour en temps réel, un contrôle d'accès sécurisé, la génération d'alertes et la production de rapports. Développé selon une approche Agile avec des retours réguliers à chaque étape, le système a été déployé en utilisant le niveau gratuit d'AWS afin de réduire les coûts. Les tests ont démontré une amélioration significative de la rapidité de traitement, de la transparence des opérations et de l'expérience utilisateur. DocuTrack Pro constitue ainsi une solution moderne, sécurisée et flexible pour la gestion documentaire au sein de toute organisation.

Mots-clés: Informatique en nuage, Système de gestion documentaire, Services AWS, Architecture serverless, Suivi en temps réel.

LIST OF FIGURES

Figure 1 :Gantt Chart Illustration Of Work Plan	3
Figure 2 : Microsoft SharePoint	5
Figure 3 : Google Drive	6
Figure 4 : Dropbox Business	7
Figure 5 : Box Enterprise	8
Figure 6 :Service Interconnection	17
Figure 7 : The Agile Model	19
Figure 8 : System Architecture	20
Figure 9 : Use Case Diagram	22
Figure 10 : Class Diagram	24
Figure 11 : Activity Diagram	25
Figure 12 : Sequence Diagram	26
Figure 13 : Relationship Diagram	34
Figure 14 : Local Deployment	35
Figure 15 : Production Deployment	36
Figure 16 : Home Page	38
Figure 17 : Sign Up Page	38
Figure 18 : Sign In Page	39
Figure 19 : Deputy Director Dashboard	39
Figure 20 : Processing Staff Dashboard	40
Figure 21 : User (Client) Dashboard	40

LIST OF TABLES

Table 1 : Functional Requirements	12
Table 2 : Non Functional Requirements	13
Table 3 : Specific Services	15
Table 4 : Frontend Technologies	28
Table 5 : Backend Technologies	29
Table 6 : Database Technologies	29
Table 7 : Libraries	30
Table 8 : Tools	31

LIST OF ABBREVIATION

ACRONYM	MEANING
ACID	Atomicity Consistency Isolation Durability
API	Application Programming Interface
AWS	Amazon Web Service
ARIA	Accessible Rich Internet Applications
CI/CD	Continuous Integration/Continuous Deployment
CDN	Content Delivery Network
CSS	Cascading Style Sheet
DLP	Data Loss Prevention
DMS	Document Management System
EC2	Elastic Compute Cloud
FR	Functional Requirement
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
HTML	Hyper Text Markup Language
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol Secure
IAM	Identification and Access Management
IT	Information Technology
iOS	Iphone Operating System
JWT	JSON Web Token
MySQL	Structured Query Language
NFR	Non Functional Requirements
PDF	Portable Document Format
RDS	Relational DataBase Service
RESTful	Representational State Transfer
RBAC	Role Based Access Control
S3	Simple Storage Service
TSX	TypeScript
UML	Unified Modelling Language
WCAG	Web Content Accessibility Guideline

TABLE OF CONTENT

CERTIFICATION	I
DEDICATION	II
ACKNOWLEDGMENTS	III
ABSTRACT	IV
RESUME	V
LIST OF FIGURES	VI
LIST OF TABLES	VII
LIST OF ABBREVIATION	VIII
GENERAL INTRODUCTION	1
I. BACKGROUND STUDY	1
II. CONTEXT STUDY	1
III. PROBLEM STATEMENT	1
IV. OBJECTIVES OF THE SYSTEM	2
A. General Objective	2
B. Specific Objectives	2
V. WORK PLAN	2
CHAPTER 1: LITERATURE REVIEW	4
1.1. INTRODUCTION	4
1.2. CURRENT STATE OF CLOUD BASED DOCUMENT MANAGEMENT SYSTEMS	4
1.3. REVIEW OF EXISTING CLOUD BASED DOCUMENT MANAGEMENT SYSTEM	4
1.3.1. KEY FEATURES	4
1.3.2. BENEFITS	8
1.3.3. CHALLENGES	9
1.4. GAPS IN EXISTING CLOUD BASED DOCUMENT MANAGEMENT SYSTEM	9
1.5. CONCLUSION	9

CHAPTER 2: ANALYSIS AND DESIGN OF A CLOUD BASED DOCUMENT MANAGEMENT SYSTEM.....	11
2.1. INTRODUCTION	11
2.2. SYSTEM ANALYSIS.....	11
2.2.1. INTRODUCTION	11
2.2.2. REQUIREMENT GATHERING	11
2.2.3. SYSTEM REQUIREMENTS	12
A. Functional Requirements	12
B. Non-Functional Requirements	13
2.3. MODEL OF DEPLOYMENT ON THE CLOUD	15
2.3.1. CLOUD SERVICE PROVIDER.....	15
2.3.2. SPECIFIC SERVICES TO BE USED ON AWS	15
2.3.3. IMPLEMENTATION WORKFLOW	16
2.3.4. SERVICE INTERCONNECTION	16
2.3.5. STEPS OF DEVELOPMENT	18
2.3.6. BENEFITS OF THE PROPOSED CLOUD SOLUTION	18
2.4. SYSTEM DEVELOPMENT METHODOLOGY	18
2.4.1. METHODOLOGY	18
2.5. SYSTEM ARCHITECTURE	19
2.6. SYSTEM DESIGN	21
2.6.1. USE CASE DIAGRAM	21
2.6.2. CLASS DIAGRAM	22
2.6.3. ACTIVITY DIAGRAM	24
2.6.4. SEQUENCE DIAGRAM	25
2.7. CONCLUSION	27
CHAPTER 3: SYSTEM IMPLEMENTATION AND TESTING	28
3.1. INTRODUCTION	28
3.2. TECHNOLOGIES, LIBRARIES AND TOOLS	28
3.2.1. TECHNOLOGIES	28
3.2.2. LIBRARIES	30
3.2.3. TOOLS	31
3.3. DATABASE DESIGN	32
3.3.1. ENTITY STRUCTURE	32
3.3.2. RELATIONSHIPS	33

3.3.3. DATA MANAGEMENT	34
3.4. DEPLOYMENT	34
3.4.1. LOCAL DEPLOYMENT	34
3.4.2. PRODUCTION DEPLOYMENT	36
3.5. IMPLEMENTATION DETAILS	36
3.5.1. FRONTEND DEVELOPMENT	36
3.5.2 BACKEND DEVELOPMENT	37
3.5.3 TESTING	37
3.6. RESULTS	37
3.6.1. RESULTS OBTAINED	37
3.7. CONCLUSION	41
GENERAL CONCLUSION	42
I. INTRODUCTION	42
II. RECALL OF WORK	42
III. ACHIEVEMENTS	42
IV. FURTHER IMPROVEMENT	43
V. CONCLUSION	43
REFERENCES	44

GENERAL INTRODUCTION

I. BACKGROUND STUDY

In today's digital age, managing documents efficiently is crucial for individuals and organizations. Traditional paper-based systems are often slow, cumbersome and prone to errors. With the rise of cloud technology, there is a growing need for systems that allow users to store, manage and retrieve documents easily from anywhere. A cloud-based document management system (DMS) can handle various file formats and images. Built on contemporary web technologies including React 18, TypeScript, and Tailwind CSS for the frontend, the system leverages Amazon Web Services (AWS) cloud infrastructure to provide scalable, secure, and efficient document management capabilities. The system architecture follows modern software engineering principles, implementing a microservices approach with clear separation of concerns between presentation, business logic, and data layers. The frontend utilizes React's component-based architecture with TypeScript for type safety, while the backend operations are handled through AWS Lambda functions, providing serverless scalability and cost-effectiveness. [1]

II. CONTEXT STUDY

In today's digital transformation era, organizations across various sectors face mounting pressure to digitize their document management processes. Traditional paper-based systems and legacy digital solutions often lack the flexibility, scalability, and collaborative features required for efficient operations. The COVID-19 pandemic further accelerated the need for remote-accessible, cloud-based solutions that enable distributed teams to collaborate effectively. The DocuTrack Pro system emerges from this context, addressing specific pain points in organizational document workflows including version control, role-based access management, real-time collaboration, and comprehensive audit trails. The system is particularly relevant for organizations requiring strict document processing workflows with multiple approval levels and compliance requirements.

III. PROBLEM STATEMENT

Organizations today face several challenges in managing documents efficiently. Manual routing and approval processes often lead to delays and workflow bottlenecks, reducing overall productivity. A lack of real-time visibility into document status results in limited transparency and accountability across teams. Inadequate security and access controls expose sensitive information to unauthorized access, posing compliance and privacy risks. Additionally, poor version control can lead to confusion and data inconsistencies, while disconnected systems hinder effective collaboration among team members. The absence of robust reporting and analytics further limits the ability to monitor performance and make informed decisions. Finally, many legacy or on-premise solutions struggle to scale with increasing document volumes and expanding user bases, making them unsuitable for growing organizational needs.

IV. OBJECTIVES OF THE SYSTEM

A. General Objective

The primary goal is to develop and implement a comprehensive cloud-based document management system that automates document processing workflows while providing robust security, real-time tracking, and advanced analytics capabilities to enhance organizational efficiency and compliance. [1]

B. Specific Objectives

1. **Authentication and Authorization:** Implement secure user authentication with role-based access control supporting three user types (Client/User, Processing Staff, Deputy Director).
2. **Document Management:** Provide comprehensive document upload, processing, version control, and lifecycle management.
3. **Workflow Automation:** Automate document routing and approval processes based on predefined business rules.
4. **Real-time Notifications:** Implement instant notification system for document status updates and deadline reminders.
5. **Reporting and Analytics:** Generate automated reports and performance analytics for decision support.
6. **Cloud Integration:** Leverage AWS free tier services for cost-effective, scalable infrastructure
7. **User Experience:** Deliver responsive, intuitive interfaces optimized for multiple devices and user roles.
8. **Data Security:** Ensure comprehensive data protection and compliance with industry standards.

V. WORK PLAN

A workplan is a detailed document or outline that describes the tasks, goals, timelines, resources, and responsibilities required to complete a specific project or activity. It serves as a roadmap that guides team members on what needs to be done, by whom, when, and how, ensuring the project stays organized, on track, and within scope.

CLOUD BASED DOCUMENT MANAGEMENT SYSTEM

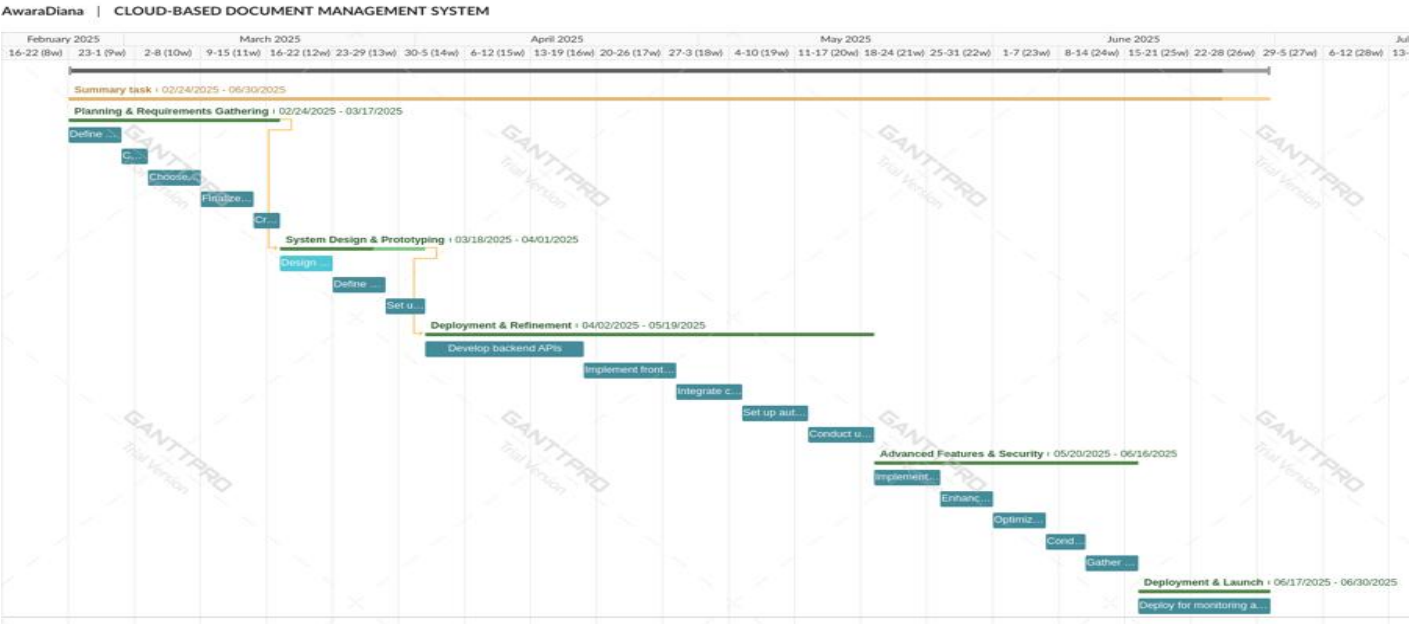


Figure 1:Gantt Chart Illustration Of Work Plan

CHAPTER 1: LITERATURE REVIEW

1.1. INTRODUCTION

The evolution of document management systems has been significantly influenced by advances in cloud computing, web technologies, and collaborative software paradigms. This literature review examines the current landscape of cloud-based document management systems, analyzing existing solutions, identifying key features and benefits, and highlighting gaps that the DocuTrack Pro system addresses. [1]

1.2. CURRENT STATE OF CLOUD BASED DOCUMENT MANAGEMENT SYSTEMS

Contemporary cloud-based document management systems have evolved from simple file storage solutions to sophisticated platforms incorporating artificial intelligence, machine learning, and advanced workflow automation. Current systems emphasize: [1]

- ❖ **Microservices Architecture:** Modern systems adopt microservices patterns enabling independent scaling and maintenance of different system components. This approach improves system resilience and allows for technology stack diversity.
- ❖ **API-First Design:** Current solutions prioritize API-driven architectures facilitating integration with third-party systems and enabling custom application development.
- ❖ **Real-time Collaboration:** WebSocket technologies and operational transformation algorithms enable simultaneous multi-user editing and real-time synchronization.
- ❖ **AI Integration:** Machine learning capabilities for content classification, optical character recognition (OCR), and intelligent search are becoming standard features.
- ❖ **Mobile-First Approach:** Responsive design and mobile applications ensure accessibility across diverse device ecosystems.

1.3. REVIEW OF EXISTING CLOUD BASED DOCUMENT MANAGEMENT SYSTEM

Several cloud-based document management systems are available today, each with unique features and capabilities. Popular options include Google Drive, Dropbox and Microsoft OneDrive. These systems typically allow users to upload and share files, collaborate in real-time and access documents from multiple devices. However, the effectiveness of these systems can vary based on their specific functionalities and user experiences. [1]

1.3.1. KEY FEATURES

1. Microsoft SharePoint Online:

SharePoint Online is a cloud-based collaboration and document management platform that is part of the Microsoft 365 suite. It allows organizations to build intranet sites, manage content,

automate workflows, and collaborate on documents with version control and co-authoring features. It's highly customizable and integrated with other Microsoft tools like Teams, OneDrive, and Power Automate.



Figure 2: Microsoft SharePoint

A. Advantages:

- ❖ Provides advanced workflow automation using Power Automate, enabling streamlined document approval and business processes.
- ❖ Fully integrates with the Microsoft 365 ecosystem, making it seamless to use alongside tools like Teams, Outlook, OneDrive, Word, and Excel.
- ❖ Offers version control and co-authoring, allowing multiple users to collaborate on documents with a clear history of changes.
- ❖ Includes robust security and compliance features, such as data loss prevention (DLP), audit logs, encryption, and regulatory compliance support (e.g., GDPR, HIPAA).

B. Limitations:

- ❖ The platform can be complex to set up and configure, especially for organizations new to Microsoft 365 or without IT support.
- ❖ Has a steep learning curve for end users due to its wide range of features and customization options.
- ❖ May be costly for small businesses, especially if only basic document management features are needed.
- ❖ Can experience performance slowdowns with large document libraries or poorly managed site structures.

2. Google Workspace (Drive):

Google Drive, a core part of Google Workspace, is a cloud storage and collaboration tool that enables real-time document editing and sharing. It integrates seamlessly with Google's productivity suite, including Docs, Sheets, Slides, and Gmail. It's known for its ease of use, smart search, and strong collaboration features, especially for distributed teams and education.



Figure 3: Google Drive

A. Advantages:

- ❖ Enables real-time collaboration in Docs, Sheets, and Slides, allowing users to work together on the same document simultaneously.
- ❖ Features intelligent search capabilities powered by Google's AI, making it fast and easy to locate documents.
- ❖ Provides seamless integration with Google's productivity suite, including Gmail, Calendar, Meet, and more.
- ❖ Supports advanced sharing and permission management, letting users set detailed access controls at the file and folder level.

B. Limitations:

- ❖ Offers limited advanced workflow automation compared to platforms like SharePoint or Box.
- ❖ Enterprise administration and compliance tools are not as extensive as those offered by Microsoft or Box.
- ❖ Users may experience compatibility and formatting issues when converting between Google Docs and Microsoft Office formats.
- ❖ Offline access and file handling may be less smooth for non-Google file types.

3. Dropbox Business:

Dropbox Business is a cloud storage and file synchronization service designed for teams and businesses. It focuses on providing a simple, intuitive interface for storing, syncing, and sharing files. Features like Smart Sync and robust integration with third-party apps make it a popular choice for small to mid-sized teams needing efficient file management without complex IT overhead.



Figure 4: Dropbox Business

A. Advantages:

- ❖ Known for simple and reliable file synchronization across devices and platforms.
- ❖ Includes Smart Sync, which allows users to access cloud-stored files locally without consuming disk space.
- ❖ Integrates with many popular business applications, such as Slack, Zoom, and Microsoft Office.
- ❖ Offers a clean and intuitive user interface, making it easy to use without extensive training.

B. Limitations:

- ❖ Lacks native document editing and real-time collaboration tools, relying instead on integrations with third-party apps.
- ❖ Workflow automation capabilities are minimal compared to SharePoint or Box.
- ❖ Version history and file recovery are limited to a certain number of days unless on higher-tier plans.
- ❖ Pricing may be high for teams that need only basic cloud storage features.

4. Box Enterprise:

Box Enterprise is a cloud content management and collaboration platform built for enterprises with strict security, compliance, and integration needs. It offers powerful admin controls, workflow automation via Box Relay, content analytics, and a developer-friendly API. It's ideal for large organizations that require secure and customizable solutions for managing and sharing content across teams.



Figure 5: Box Enterprise

A. Advantages:

- ❖ Provides enterprise-grade security, including encryption, access controls, audit logs, and compliance with standards like HIPAA and GDPR.
- ❖ Supports workflow automation through Box Relay, allowing for customizable approval processes and task flows.
- ❖ Offers content analytics, giving insights into file access, collaboration, and activity patterns.
- ❖ Features an API-driven architecture, enabling businesses to build custom apps and integrations on top of the Box platform.

B. Limitations:

- ❖ The platform can be complex and require technical expertise to fully utilize its advanced features.
- ❖ Has no native document editing tools; relies on integrations with Google Docs, Office 365, or other platforms.
- ❖ Pricing can be high, especially for smaller teams or companies that don't need extensive enterprise features.
- ❖ Some features may be overkill for basic use cases, such as simple file sharing or storage.

1.3.2. BENEFITS

- ❖ **Cost Reduction:** Elimination of hardware investment and maintenance costs
- ❖ **Scalability:** Automatic scaling based on demand without infrastructure management
- ❖ **Accessibility:** Global access from any device with internet connectivity

- ❖ **Collaboration:** Enhanced team productivity through real-time collaboration features
- ❖ **Security:** Enterprise-grade security measures including encryption and access controls
- ❖ **Disaster Recovery:** Automatic backup and recovery capabilities
- ❖ **Compliance:** Built-in compliance tools for regulatory requirements

1.3.3. CHALLENGES

Despite their advantages, cloud-based document management systems face several challenges:

- ❖ **Data Privacy Concerns:** Organizations worry about data control and location
- ❖ **Internet Dependency:** Reduced functionality during connectivity issues
- ❖ **Integration Complexity:** Challenges in integrating with legacy systems
- ❖ **Vendor Lock-in:** Difficulty in migrating between platforms
- ❖ **Customization Limitations:** Restricted ability to modify core functionalities
- ❖ **Performance Issues:** Latency concerns for large file operations

1.4. GAPS IN EXISTING CLOUD BASED DOCUMENT MANAGEMENT SYSTEM

Analysis of current solutions reveals several gaps that the DocuTrack Pro system addresses:

- ❖ **Role-Specific Workflow Design:** Most systems offer generic workflows rather than role-optimized interfaces and processes
- ❖ **Cost-Effective Entry Point:** Many enterprise solutions have high entry costs prohibitive for smaller organizations
- ❖ **Simplified AWS Integration:** Limited options for organizations wanting to leverage AWS free tier services effectively
- ❖ **Processing-Focused Workflows:** Lack of systems specifically designed for document processing workflows with multiple approval stages
- ❖ **Transparent Pricing:** Complex pricing models make cost prediction difficult
- ❖ **Customizable Notification Systems:** Limited flexibility in notification configuration and delivery method.

1.5. CONCLUSION

In conclusion, while existing cloud-based document management systems like Microsoft SharePoint, Google Drive, Dropbox Business, and Box Enterprise offer a range of powerful features such as real-time collaboration, integration capabilities, and enterprise-grade security, they also present notable limitations including high costs, complexity, limited workflow customization, and challenges

with integration and scalability. These systems often lack tailored solutions for specific user roles and streamlined processing workflows, especially for organizations seeking cost-effective and AWS-integrated platforms. DocuTrack Pro addresses these gaps by providing a role-specific, scalable, and affordable document management solution built on modern web technologies and a serverless AWS architecture. It enhances collaboration, ensures secure document processing, and offers flexibility in notification and workflow configuration, making it a suitable choice for organizations of all sizes aiming to improve their document handling processes. [4]

CHAPTER 2: ANALYSIS AND DESIGN OF A CLOUD BASED DOCUMENT MANAGEMENT SYSTEM

2.1. INTRODUCTION

The system analysis and design phase establishes the foundation for developing DocuTrack Pro by defining requirements, architectural decisions, and implementation strategies. This process integrates principles from software engineering, cloud computing, and user experience design to create a secure, scalable, and user-friendly platform accessible via the internet. This chapter outlines the comprehensive analysis process, system requirements, and cloud deployment model.

2.2. SYSTEM ANALYSIS

2.2.1. INTRODUCTION

When creating a cloud-based document management system, it is important to define the key system requirements. These requirements help ensure that the system meets the needs of its users and functions properly. They include both hardware and software needs, as well as user expectations, and technology assessment to ensure the solution meets organizational needs while leveraging modern cloud capabilities. The analysis focused on understanding user roles, document processing workflows, and technical constraints. [4]

2.2.2. REQUIREMENT GATHERING

Requirements were gathered through:

- ❖ **Stakeholder Interviews:** Sessions with potential users from different organizational roles.
- ❖ **Workflow Analysis:** Mapping existing document processing procedures.
- ❖ **Technology Assessment:** Evaluating available cloud services and development frameworks.
- ❖ **Competitive Analysis:** Studying existing solutions to identify improvement opportunities.

Requirements gathered:

- ❖ **User Needs:** Identify what users want from the system. This includes features like uploading files, searching documents, and sharing with others.
- ❖ **Performance Requirements:** Determine how fast the system should be. Users expect quick loading times and smooth operation, even with many documents.
- ❖ **Security Requirements:** Define how to protect user data. This includes encryption, secure login, and regular backups to prevent data loss.

- ❖ **Compatibility:** Ensure the system works on different devices like computers, tablets, and smartphones. It should also be compatible with various web browsers.
- ❖ **Storage Capacity:** Decide how much storage space is needed for users. The system should handle a large number of files without slowing down.
- ❖ **User Interface:** Gather requirements for a simple and easy-to-use interface. Users should be able to navigate the system without difficulty.
- ❖ **Support and Maintenance:** Plan for ongoing support for users and regular updates to keep the system running smoothly.

2.2.3. SYSTEM REQUIREMENTS

A. Functional Requirements

Table 1: Functional Requirements

Type	ID	Description
User Management	FR 1	System shall support user registration with role assignment
	FR 2	System shall authenticate users using email and password
	FR 3	System shall provide role-based dashboard access (Client, Processing Staff, Deputy Director)
	FR 4	System shall maintain user session management
Document Management	FR 5	System shall allow document upload in multiple formats (PDF, DOC, DOCX, TXT)
	FR 6	System shall provide document version control
	FR 7	System shall enable document status tracking (assigned, in-progress, completed)
	FR 8	System shall support document metadata

		management
	FR 9	System shall provide document search and filtering capabilities
Workflow Management	FR 10	System shall allow assigning documents to processing staff
	FR 11	System shall allow tracking document processing timelines
	FR 12	System shall allow deadline management
	FR 13	System shall provide approval workflows for deputy directors
Notification System	FR 14	System shall send real-time notifications for status changes
	FR 15	System shall provide deadline reminder notifications
	FR 16	System shall maintain notification history
Reporting and Analytics	FR 17	System shall provide analytics dashboards
	FR 18	System shall generate processing performance reports
	FR 19	System shall export reports in PDF format

B. Non-Functional Requirements

Table 2: Non Functional Requirements

Type	ID	Description
Performance	NFR 1	System shall load pages quickly

	NFR 2	System shall support concurrent access
	NFR 3	System shall handle file uploads up to 50MB
Security	NFR 4	System shall encrypt all data transmissions using HTTPS
	NFR 5	System shall implement JWT-based authentication
	NFR 6	System shall provide audit trails for all document operations
	NFR 7	System shall implement role-based access control
Usability	NFR 8	System shall maintain consistent user interface across all modules
	NFR 9	System shall provide intuitive navigation with minimal training required
Reliability	NFR 10	System shall maintain 99.5% uptime
	NFR 11	System shall implement graceful error handling
Scalability	NFR 12	System shall automatically scale based on user demand.
	NFR 13	System shall support horizontal scaling of processing capabilities

2.3. MODEL OF DEPLOYMENT ON THE CLOUD

2.3.1. CLOUD SERVICE PROVIDER

For this cloud-based document management system, Amazon Web Services (AWS) is chosen as the cloud service provider. For the cloud-based document management system, the public cloud deployment model is chosen. This means the system will be available to all users over the internet, hosted on a public cloud service provider like Amazon Web Services (AWS). Using a public cloud deployment model allows for easy access, cost-effectiveness, and scalability, enabling users to manage their documents without worrying about hardware or infrastructure. Below are the reasons for my choice:

- ❖ **Scalability:** AWS can easily grow with the system's needs, allowing more storage and computing power as required.
- ❖ **Security:** AWS offers strong security features to protect user data, including encryption and access controls.
- ❖ **Cost-Effectiveness:** AWS has a pay-as-you-go pricing model, which is helpful for managing costs as the system grows.
- ❖ **Wide Range of Services:** AWS provides various services that are useful for building and managing the document management system.
- ❖ **Integration:** Comprehensive service ecosystem enables seamless integration.
- ❖ **Performance:** Global CDN and edge locations ensure optimal performance.

2.3.2. SPECIFIC SERVICES TO BE USED ON AWS

Table 3: Specific Services

Service	Logo	Name	Usage
Compute Services		AWS Lambda	Serverless functions for backend processing
		API Gateway	RESTful API management and routing
Storage Services		Amazon S3	Object storage for documents and static assets

CLOUD BASED DOCUMENT MANAGEMENT SYSTEM

		Amazon RDS (MySQL)	Relational database for structured data
Security Services	 AWS Identity and Access Management (IAM)	AWS IAM:	Identity and access management
Monitoring and Management		CloudWatch	Application monitoring and logging.

2.3.3. IMPLEMENTATION WORKFLOW

- ❖ **Infrastructure Setup:** Deploy AWS resources
- ❖ **Database Initialization:** Execute MySQL schema creation scripts
- ❖ **Lambda Function Deployment:** Deploy backend processing functions
- ❖ **API Gateway Configuration:** Configure REST endpoints and authentication.
- ❖ **Frontend Deployment:** Deploy React application to S3 .
- ❖ **Security Configuration:** Implement IAM policies and security groups.
- ❖ **Monitoring Setup:** Configure CloudWatch alarms and dashboards.

2.3.4. SERVICE INTERCONNECTION

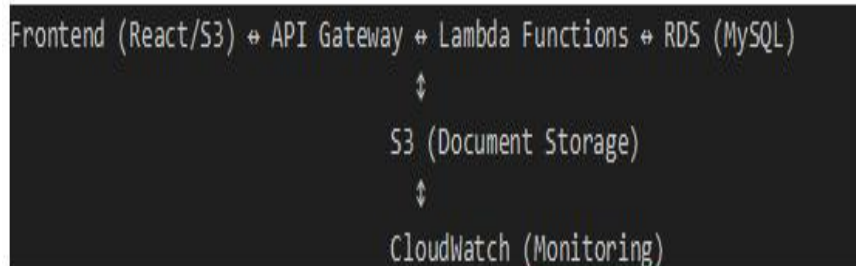


Figure 6:Service Interconnection

The serverless cloud architecture of the DocuTrack Pro system is designed for high availability, scalability, and cost-effectiveness while ensuring enterprise-grade security and performance. At the frontend layer, a React application built with TypeScript is hosted as static files on Amazon S3, which serves the compiled UI directly to users' browsers. The frontend communicates bidirectionally with the backend through secure HTTP/HTTPS protocols.

At the API Gateway layer, Amazon API Gateway acts as the central hub for all client requests. It routes incoming traffic to the appropriate Lambda functions based on the defined endpoints. It also enforces security policies such as authentication (via JWT), CORS, and rate limiting, and manages responses returned to the client.

The Lambda functions layer provides the serverless compute power behind the system. Each Lambda function handles a specific piece of business logic, such as authentication, document handling, or notifications. These functions are stateless, executing only when triggered by an API call, which ensures scalability and cost efficiency by charging only for actual compute time used.

The RDS (MySQL) database layer serves as the structured data store. It holds user accounts, document metadata, processing logs, and notifications. RDS ensures ACID compliance for data consistency and supports relational integrity through foreign key constraints. It also provides built-in backup and recovery features for data protection.

For file management, the system uses a separate S3 bucket for document storage. This bucket stores PDFs, Word documents, text files, and more, with virtually unlimited capacity. S3 ensures security through encryption at rest and in transit and enforces access control policies. Lambda functions interface with S3 to handle uploads, retrievals, and deletions.

Amazon CloudWatch is integrated for monitoring and observability. It collects logs from Lambda functions, API Gateway, and other services, aggregates system metrics, and provides alerting mechanisms when performance thresholds are exceeded or errors occur. This enables ongoing performance optimization and system health monitoring.

The data flow begins when the user sends a request from the frontend to API Gateway. API Gateway authenticates the request using JWT tokens, routes it to the appropriate Lambda function, which then interacts with either the RDS database or S3 storage as needed. CloudWatch logs every step, and the processed response flows back through API Gateway to the frontend. This architecture supports dynamic

workloads, ensures reliability, and reduces operational overhead by leveraging fully managed AWS services.

2.3.5. STEPS OF DEVELOPMENT

- ❖ **Foundation:** Setup development environment and AWS infrastructure
- ❖ **Authentication:** Implement user management and authentication
- ❖ **Core Features:** Develop document management and workflow capabilities
- ❖ **Advanced Features:** Implement notifications and reporting
- ❖ **Testing:** Comprehensive testing and performance optimization
- ❖ **Deployment:** Production deployment and monitoring setup

2.3.6. BENEFITS OF THE PROPOSED CLOUD SOLUTION

- ❖ **Reduced Infrastructure Costs:** No upfront hardware investment
- ❖ **Automatic Scaling:** Handles traffic spikes without manual intervention
- ❖ **Global Accessibility:** Available worldwide with optimal performance
- ❖ **Enhanced Security:** Enterprise-grade security without additional investment
- ❖ **Rapid Development:** Managed services accelerate development timeline
- ❖ **Compliance:** AWS compliance certifications support regulatory requirements

2.4. SYSTEM DEVELOPMENT METHODOLOGY

2.4.1. METHODOLOGY

Methodology is the theoretical or systematic analysis of data or methods that are being applied in the field of study. It consist of theoretical analysis of methods and principles in the branch of knowledge. This section outlines the system's requirement and specification, the methodologies employed, the materials utilized, and the software chosen for system diagramming.

Addressing the rigidity of traditional models, the Agile approach develops software through iterative cycles (sprints), enabling continuous testing and adaptation. Unlike linear methods, Agile allows teams to incorporate feedback and refine requirements at each stage, ensuring faster delivery of functional components while maintaining flexibility for changes. The Agile model is a flexible and iterative approach to software development that emphasizes collaboration, customer feedback, and rapid delivery of functional software.



Figure 7: The Agile Model

This section outlines how the Agile methodology can be applied to the development of a cloud-based document management system. Here is a detailed explanation of how to implement a cloud based document management system using the agile model

1. **Planning:** The team meets to discuss the project goals and create a list of features needed. This list is called the "backlog."
2. **Design:** The team designs how the software will look and work. They decide on the main features to build first.
3. **Development:** The team works on the first set of features in a short period (sprint). They write code and create the software.
4. **Testing:** After development, the team tests the features to ensure they work correctly and are free of bugs.
5. **Review:** At the end of the sprint, the team shows the completed features to users and gathers feedback.
6. **Adjust:** Based on feedback, the team makes changes or adds new features to the backlog for the next sprint.
7. **Repeat:** The process starts over with the next sprint, allowing the team to continue improving the software until it meets user needs.

2.5. SYSTEM ARCHITECTURE

The DocuTrack Pro system employs a serverless microservices architecture built on AWS cloud infrastructure, ensuring optimal performance and cost efficiency while maintaining enterprise-grade security standards.



Figure 8: System Architecture

The figure above illustrates a cloud-based, serverless system architecture composed of three main layers: the Presentation Layer, the Application Layer, and the AWS Cloud Layer. At the top, the Presentation Layer represents the user interface built with React and TypeScript (TSX). It includes three distinct dashboards tailored to different user roles: the User Dashboard, designed for general users to access their documents and notifications; the Staff Dashboard, intended for processing staff to manage and handle documents; and the Director Dashboard, which provides high-level access to reports, monitoring tools, and administrative functionalities. These dashboards communicate with backend services via API calls.

The middle Application Layer contains the system's core functionality. It consists of modular services such as the Authentication Service for managing user login and identity, the Document Service responsible for document creation, retrieval, and storage logic, and the Notification Service, which handles the delivery of alerts and status updates to users. These services are deployed as AWS APIs, ensuring scalability and modular design.

At the base is the AWS Cloud Layer, which provides the infrastructure and backend services that support the application. API Gateway acts as the secure entry point for all frontend requests, while Lambda Functions execute backend logic in response to those requests. RDS is used as the primary MySQL database to store user and document data. Amazon S3 handles file storage for documents and assets.

Finally, CloudWatch provides monitoring, logging, and performance tracking to ensure operational reliability and observability.

Overall, this architecture showcases a scalable, secure, and modular cloud solution leveraging AWS services and modern frontend technologies to support document tracking and role-based interaction.

2.6. SYSTEM DESIGN

2.6.1. USE CASE DIAGRAM

A use case diagram for a cloud-based document management system (DMS) visually represents the interactions between users (actors) and the system, highlighting the functional requirements and major functionalities. (Scribd, 2014)

2.6.1.1. DESCRIPTION OF THE USE CASE DIAGRAM

The Use Case Diagram for DocuTrack Pro illustrates the system's functionality from the user's perspective:

❖ Primary Actors:

Client/User: Submits documents, tracks progress, receives notifications

Processing Staff: Processes documents, updates status, manages workload

Deputy Director: Oversees operations, generates reports, manages users

❖ Key Use Cases:

Document Submission: Clients upload documents with metadata

Document Processing: Staff members edit and process documents

Status Tracking: Real-time monitoring of document progress

Notification Management: Automated alerts for status changes

Report Generation: Analytics and performance reports

User Management: Role-based access control and user administration

System Boundary: The web-based application with AWS cloud services

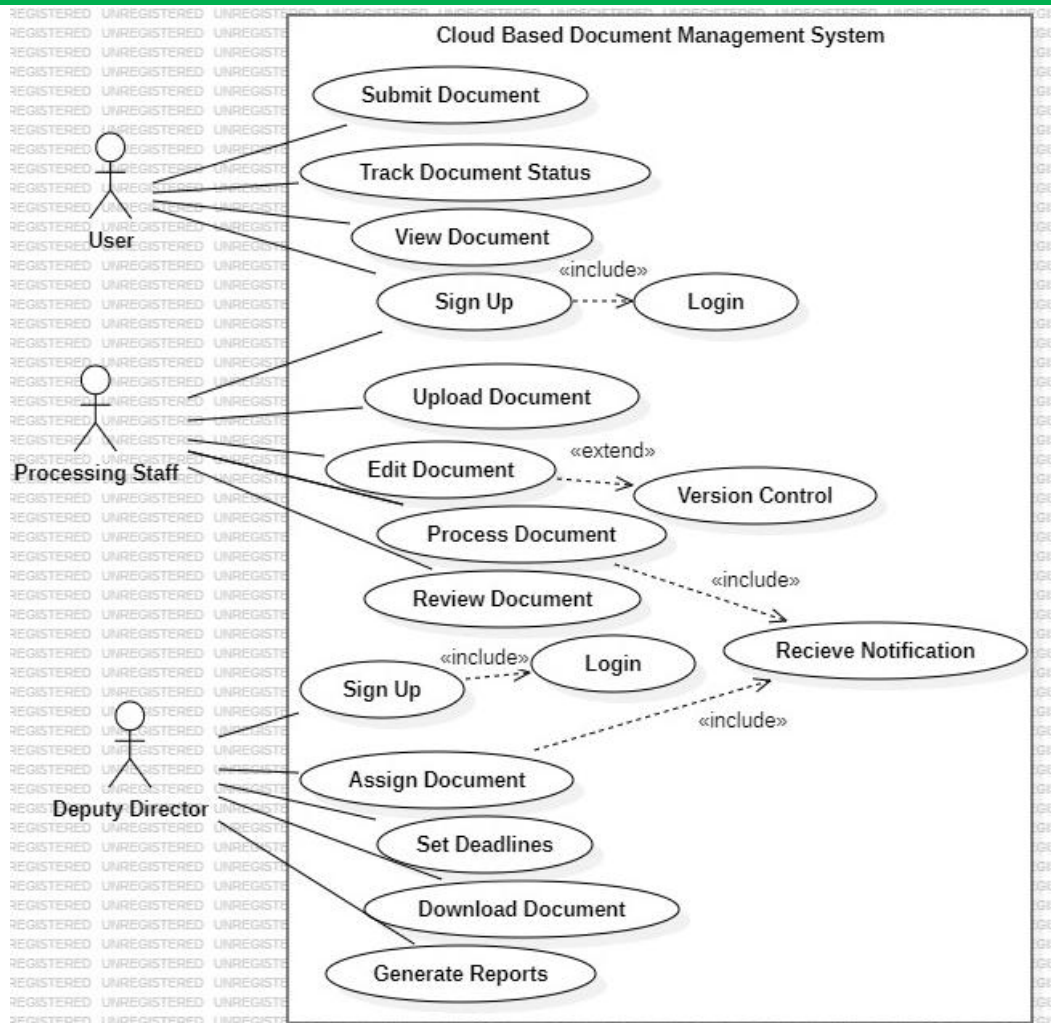


Figure 9: Use Case Diagram

2.6.2. CLASS DIAGRAM

A class diagram is a type of static structure diagram that represents the classes, attributes, methods, and relationships within a system. In the context of a cloud-based document management system (DMS), a class diagram represents the object-oriented structure of the system. (Scribd, 2014)

2.6.2.1. DESCRIPTION OF THE CLASS DIAGRAM

❖ Core Classes:

User Class: Attributes (id, email, role, name) and methods (authenticate, updateProfile)

Document Class: Attributes (id, name, content, status, priority) and methods (upload, update, download)

Notification Class: Attributes (type, message, timestamp) and methods (send, markRead)

Report Class: Attributes (type, data, format) and methods (generate, export)

❖ Service Classes:

AuthenticationService: Manages user login and authorization

DocumentService: Handles document operations and file management

NotificationService: Manages system notifications and alerts

ReportService: Generates analytics and performance reports

- ❖ **Relationships:** Inheritance, composition, and association relationships between classes showing system structure

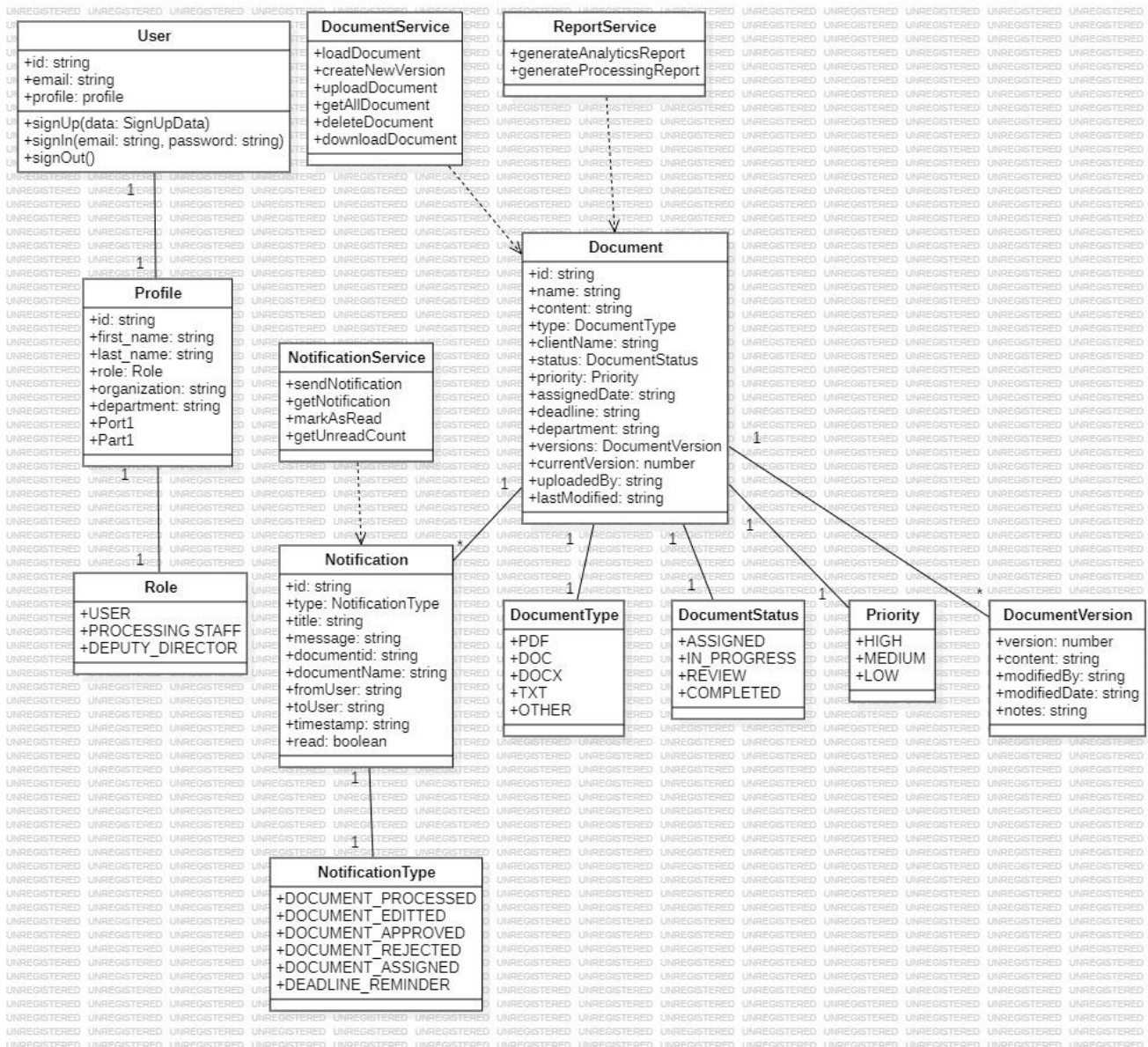


Figure 10: Class Diagram

2.6.3. ACTIVITY DIAGRAM

An activity diagram is a diagram that illustrates the workflow processes within the system. The activity diagram below illustrates the workflow of the document management process in DocuTrack Pro, detailing how users interact with the system. The process begins when a user logs into the platform, after which they can choose to upload a document. Once a document is uploaded, it goes through a validation phase, checking for file format, metadata, and any required fields. If the document is valid, it is forwarded to the processing staff, who are responsible for reviewing and updating its status. After processing, the document is either sent for approval to the deputy director or marked as completed, depending on its priority and content. At each step, the system sends out notifications to relevant users to ensure real-time updates and communication. This diagram effectively maps out a structured, role-based flow that ensures accountability, proper routing, and timely feedback in document handling.

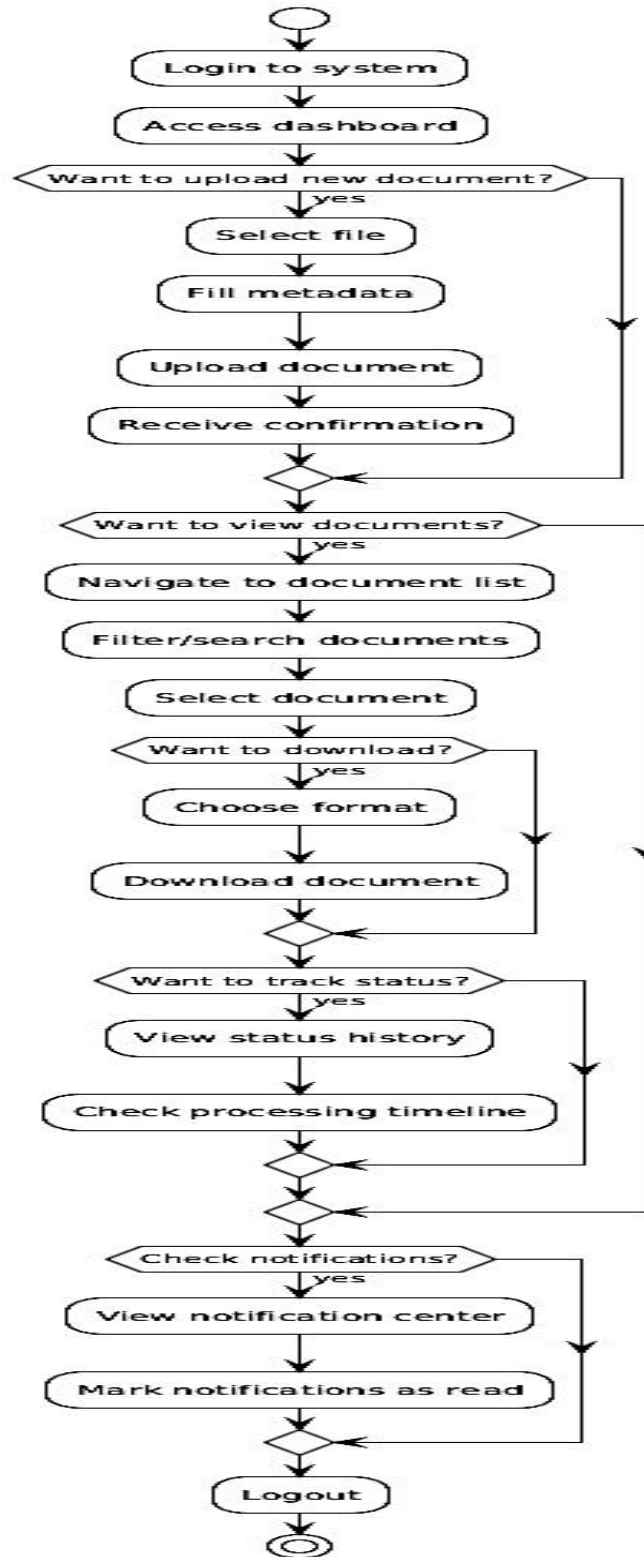


Figure 11:Activity Diagram

2.6.4. SEQUENCE DIAGRAM

A sequence diagram is a type of interaction diagram that illustrates how objects interact in a particular scenario of a system over time. It shows the sequence of messages exchanged between objects, focusing on the order in which these interactions occur. In a cloud-based document

management system (DMS), a sequence diagram explains the flow of operations for specific use cases. The diagram below explains the use case of document submission.

The sequence diagram describes the interaction between system components during a document submission and approval process. It starts with the client initiating the upload, sending the document and metadata to the system. The system first authenticates the client and then saves the document to Amazon S3 for storage while metadata is recorded in MySQL (RDS). Next, the system triggers a notification to the processing staff, who retrieve and review the document. Once reviewed, the staff member updates the status, and a second notification is sent to the deputy director for approval. The deputy director then either approves or rejects the document, and the final decision is stored back in the database. Throughout the process, each action is tracked in the system logs, and users receive feedback via notifications. This diagram emphasizes the chronological order of interactions and highlights the importance of system components like authentication, storage, and notifications in facilitating a smooth document workflow.

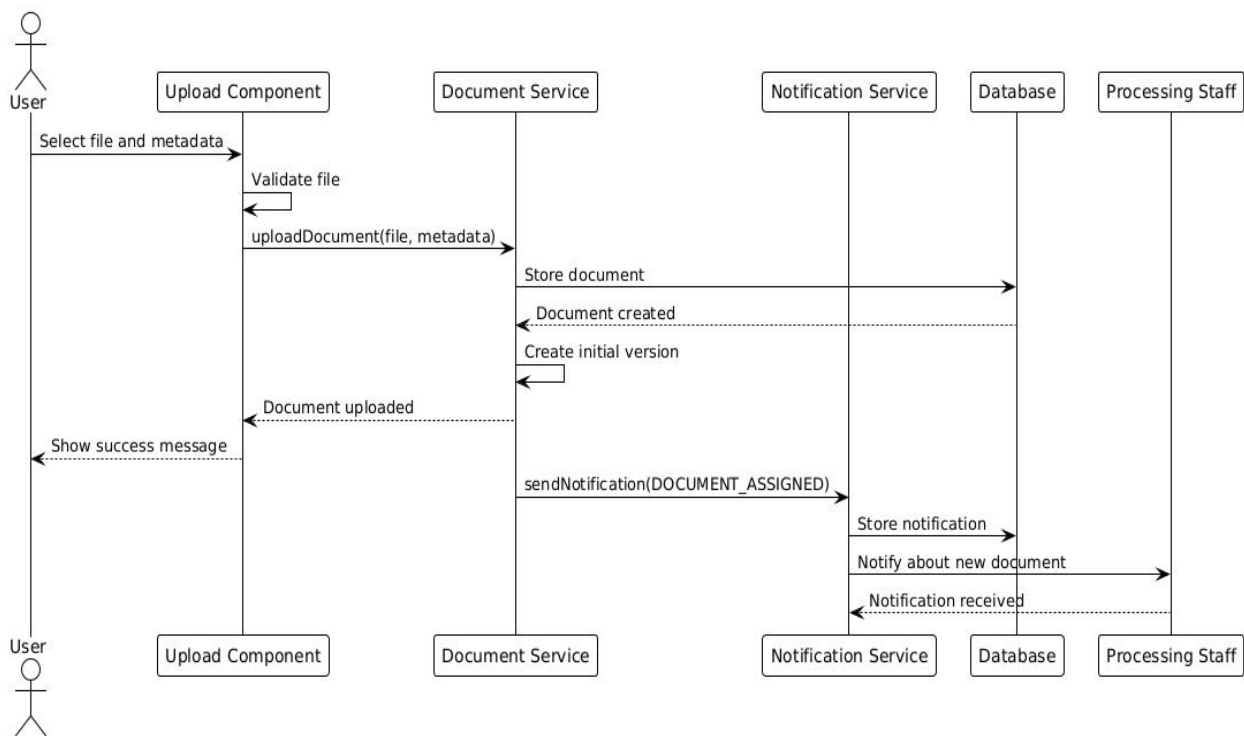


Figure 12:Sequence Diagram

2.7. CONCLUSION

The UML diagrams provide a comprehensive view of the DocuTrack Pro system architecture and functionality. The Use Case Diagram identifies user interactions, the Class Diagram shows the object-oriented structure, the Activity Diagram illustrates workflows, and the Sequence Diagram demonstrates component interactions. Together, these diagrams form a complete technical specification that guided the system implementation and serves as documentation for future maintenance and enhancements.

CHAPTER 3: SYSTEM IMPLEMENTATION AND TESTING

3.1. INTRODUCTION

This chapter presents the implementation details and testing procedures of DocuTrack Pro, a comprehensive cloud-based document management system. The system was developed using modern web technologies and deployed on Amazon Web Services (AWS) infrastructure to ensure scalability, reliability, and cost-effectiveness. The implementation follows industry best practices for security, user experience, and system architecture.

3.2. TECHNOLOGIES, LIBRARIES AND TOOLS

3.2.1. TECHNOLOGIES

Technology refers to the broad set of methods, practices, processes, and systems developed to solve problems and perform specific tasks. It encompasses a wide range of applications, including software, hardware and methodologies. Here's an overview of the key technologies involved:

❖ Frontend Technologies:

Table 4: Frontend Technologies

Technology	Icon	Description
React 18		Modern JavaScript library for building user interfaces with component-based architecture
TypeScript		Strongly typed programming language that builds on JavaScript, providing better code quality and developer experience

Vite		Next-generation frontend build tool providing fast development server and optimized builds
HTML5/CSS3		Standard markup and styling languages for web development

❖ **Backend Technologies:***Table 5: Backend Technologies*

Technology	Icon	Description
AWS Lambda		Serverless compute service for running backend logic without managing servers
AWS S3 Bucket		Scalable cloud storage container used to store and manage data such as files, documents, images, videos, and backups
Node.js		JavaScript runtime environment for server-side development
API Gateway		Fully managed service for creating, publishing, and managing APIs

❖ **Database Technologies:***Table 6: Database Technologies*

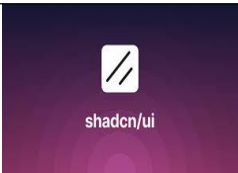
Technology	Icon	Description
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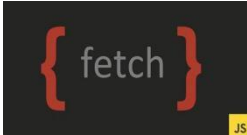
MySQL (RDS)		Relational database management system for structured data storage
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3.2.2. LIBRARIES

A library is a collection of pre-written code, functions, and routines that developers can use to perform common tasks without writing code from scratch. Libraries provide functionalities and abstractions to simplify the development process.

Table 7:Libraries

Library	Icon	Description
Shadcn/UI		Modern component library built on Radix UI primitives
Tailwind CSS		Utility-first CSS framework for rapid UI development
Lucide React		Beautiful and customizable SVG icons for React applications
TanStack Query (@tanstack/react-query):		Powerful data synchronization library for React applications
React Router DOM		Declarative routing library for React web applications
React Hook Form		Performant forms library with easy validation

Zod		TypeScript-first schema declaration and validation library
JWT (jsonwebtoken)		Used for implementing secure authentication and authorization via JSON Web Tokens.
Axios/Fetch		For making HTTP requests between the frontend and backend.

3.2.3. TOOLS

A tool is a software application or utility designed to perform specific, often repetitive tasks to assist developers in their work. Tools can be standalone applications or integrated into larger systems to enhance productivity and efficiency. Below is an introduction to some key tools used:

Table 8: Tools

Tool	Icons	Description
Visual Studio Code		The primary code editor used for development, offering rich extensions and debugging capabilities.
AWS		Offers a wide range of on-demand services, such as computing power, storage, databases, and more.
GitHub		A web-based platform for version control and collaboration, built around Git, a distributed version control system.
Postman		Used for testing API endpoints and verifying backend logic.

Npm		Package managers for installing and managing project dependencies.
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3.3. DATABASE DESIGN

A well-structured database is fundamental to the operation of the platform. The platform uses a relational database, with the schema defined and managed using MySQL. The database design ensures data integrity, supports complex relationships, and enables efficient querying for analytics and reporting.

3.3.1. ENTITY STRUCTURE

The system utilizes MySQL for document storage and structured data:

❖ User Entity:

id (Primary Key)
email (Unique)
password_hash
first_name, last_name
role (user, processing-staff, deputy-director)
organization, department
created_at, updated_at

❖ Document Entity:

id (Primary Key)
name, content
file_type (pdf, doc, docx, txt, other)
client_name, status
priority (low, medium, high)
assigned_date, deadline
department, current_version
uploaded_by_id, s3_file_key

❖ Notification Entity:

id (Primary Key)
type, title, message

document_id, document_name
 from_user_id, to_user_id
 is_read, created_at

3.3.2. RELATIONSHIPS

- ❖ One-to-Many: User → Documents (uploaded_by relationship)
- ❖ One-to-Many: Document → Document_Versions
- ❖ One-to-Many: User → Notifications (both sender and receiver)
- ❖ Many-to-One: Notifications → Documents
- ❖ One-to-Many: Document → Processing_Logs

First, there is a one-to-many relationship between users and documents. A single user whether a client, processing staff, or deputy director can upload multiple documents over time, but each document is linked to only one uploader through the `uploaded_by_id` foreign key. This association is critical for tracking document ownership, enforcing access controls, and generating user-specific reports. For example, a client may upload ten different documents, and each will be tied directly to that user's account.

Next, a one-to-many relationship exists between documents and document versions. As documents go through revisions during the workflow, new versions are created. Each version captures the full content state at a particular point in time and is stored in the `document_versions` table, along with metadata like who made the changes and when. This enables version control, change tracking, and the ability to revert to earlier versions. For instance, a contract might have five versions during its review and approval process.

The system also models two separate one-to-many relationships between users and notifications as both senders and receivers. A user can send multiple notifications (`from_user_id`), and each notification identifies its sender for auditing purposes. Simultaneously, a user can receive numerous notifications (`to_user_id`), which form their personal notification inbox with read/unread statuses. This dual association supports efficient communication tracking and allows users to manage their message history effectively.

Furthermore, there is a many-to-one relationship between notifications and documents. A single document can generate multiple notifications over its lifecycle such as upload confirmations, status updates, review assignments, and completion alerts but each notification

references only one specific document. This link supports document-focused filtering and ensures users stay informed about activities related to the documents they are involved in. For example, one document might produce eight notifications at various workflow stages.

Lastly, a one-to-many relationship exists between documents and processing logs. Every significant action taken on a document such as edits, status updates, or user assignments creates a new log entry. These logs provide a comprehensive history of the document's journey and are essential for auditing, compliance, troubleshooting, and performance analysis. For example, a single document might accumulate fifteen log entries as it progresses from upload to final approval.

Together, these relationships create a robust data model that supports secure, traceable, and efficient document management throughout the entire lifecycle.

3.3.3. DATA MANAGEMENT

❖ ENTITY RELATIONSHIP DIAGRAM

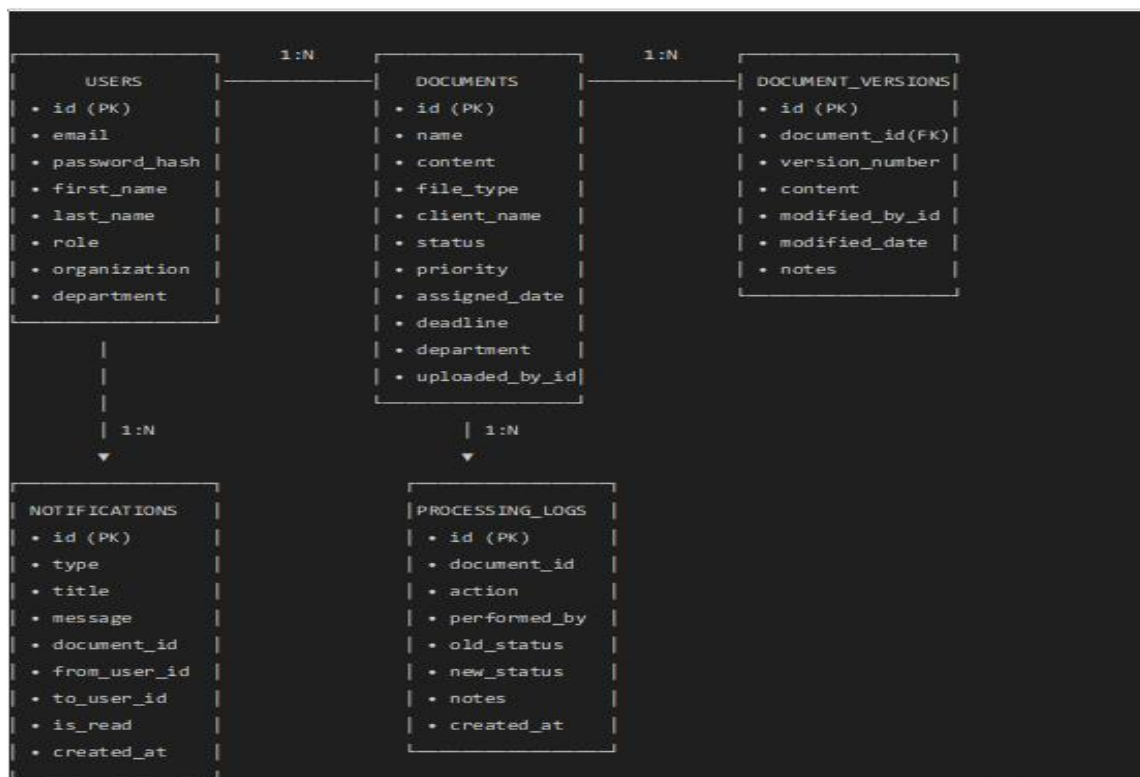


Figure 13: Relationship Diagram

3.4. DEPLOYMENT

3.4.1. LOCAL DEPLOYMENT

The local development environment is configured using:

- ❖ **Vite Development Server:** Hot module replacement for rapid development
- ❖ **Environment Variables:** Local configuration for AWS services
- ❖ **Mock Services:** Local alternatives for AWS services during development
- ❖ **Development Database:** Local MySQL instance for testing

The local development environment for DocuTrack Pro is set up to closely mimic the production environment while providing a fast, flexible, and isolated setting for building and testing features. At the core of this setup is the Vite development server, which offers hot module replacement to ensure that changes made to the frontend code are reflected instantly in the browser without a full reload, significantly speeding up the development process. To simulate real-world conditions, environment variables are used to manage sensitive configurations such as database credentials and API keys, allowing seamless switching between local and production environments. During development, mock services are employed as stand-ins for actual AWS components like Lambda or S3, enabling developers to test logic without needing to deploy to the cloud. Additionally, a local MySQL database instance is used to replicate the production database structure, allowing thorough testing of data-related operations such as user registration, document uploads, and notification handling. This environment provides developers with the tools to work efficiently, test robustly, and maintain consistency throughout the development cycle.

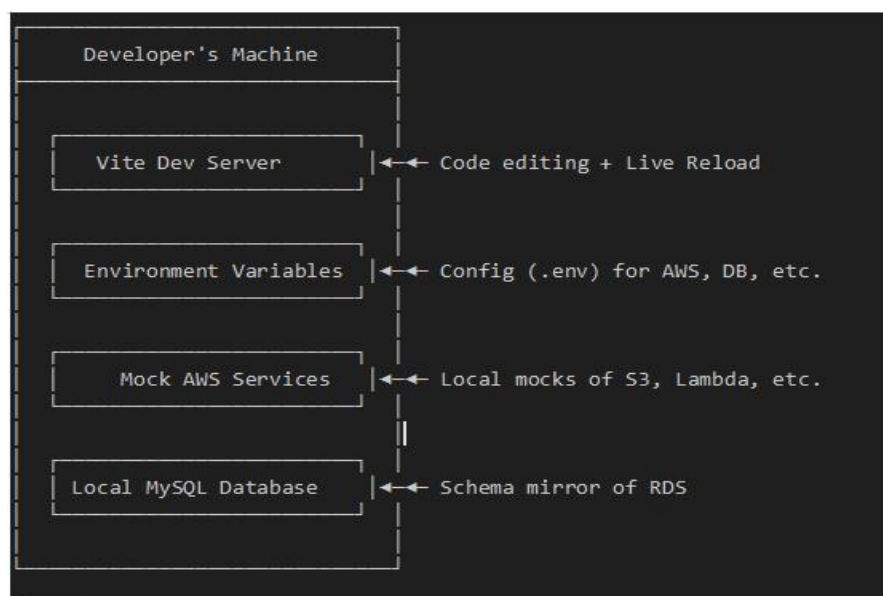


Figure 14:Local Deployment

3.4.2. PRODUCTION DEPLOYMENT

Production deployment utilizes AWS cloud infrastructure:

❖ **AWS S3:** Static website hosting for React application

In the production environment, DocuTrack Pro is deployed using AWS cloud infrastructure to ensure scalability, high availability, and cost-efficiency. One of the key components of this setup is Amazon S3 (Simple Storage Service), which is used to host the React frontend application as a static website. By uploading the compiled production build of the React app to an S3 bucket configured for static hosting, the system can deliver fast and reliable web content to users globally. S3's integration with CloudFront (if enabled) further enhances content delivery by caching assets at edge locations, reducing latency. This serverless approach to frontend hosting eliminates the need to manage web servers, simplifies deployment, and ensures seamless scaling based on user demand. It also provides robust availability and durability, making it an ideal choice for hosting static frontend assets in production.

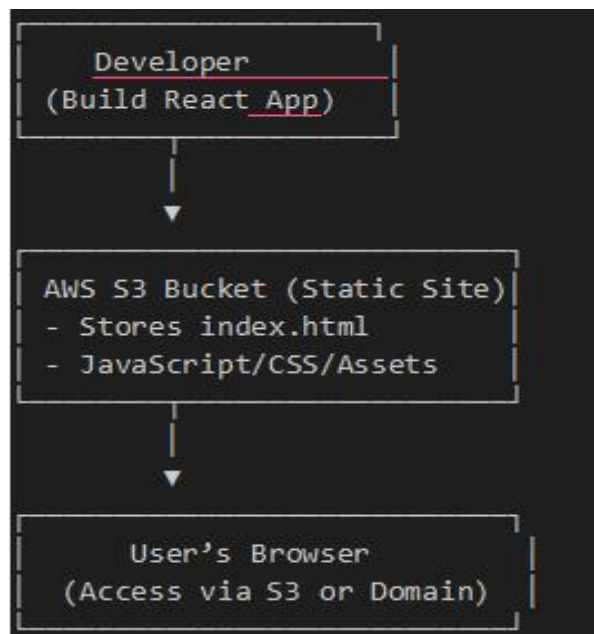


Figure 15: Production Deployment

3.5. IMPLEMENTATION DETAILS

3.5.1. FRONTEND DEVELOPMENT

The frontend of the system was developed using React with TypeScript, prioritizing modularity, maintainability, and performance. A component-based architecture was adopted, promoting reusable and self-contained components with a clear separation of concerns. State management was handled using the Context API for global state and React Query for efficient server state handling and data fetching. The user interface was designed with a mobile-first approach using Tailwind CSS, ensuring responsiveness across devices. Accessibility was a key focus, with compliance to WCAG 2.1 standards, including the use

of ARIA labels and full keyboard navigation support. To optimize performance, techniques such as code splitting, lazy loading, and bundle size optimization were implemented to enhance load times and user experience

3.5.2 BACKEND DEVELOPMENT

The backend was implemented using an AWS serverless architecture, leveraging managed services for scalability and efficiency. AWS Lambda was used to implement individual functions for each API endpoint, ensuring event-driven and stateless execution. Amazon API Gateway served as the RESTful interface, properly handling HTTP methods and status codes in accordance with best practices. JWT-based authentication was used to securely manage user sessions. All incoming data was subjected to input sanitization and validation to maintain data integrity and prevent injection attacks. The system included robust error handling mechanisms, with detailed logging and user-friendly error messages to improve debugging and user experience.

3.5.3 TESTING

A comprehensive testing strategy was implemented to ensure the reliability, security, and performance of the system. Unit testing was conducted using Jest to verify individual components and functions. Integration testing covered API endpoints using mock data to ensure smooth communication between system layers. End-to-end testing was carried out with Cypress, simulating full user workflows from login to document submission. Performance testing was also performed to evaluate the responsiveness and scalability of API endpoints and frontend performance under load. Finally, security testing focused on verifying proper authentication, authorization flows, and input validation to protect against common vulnerabilities.

3.6. RESULTS

3.6.1. RESULTS OBTAINED

❖ Home Page

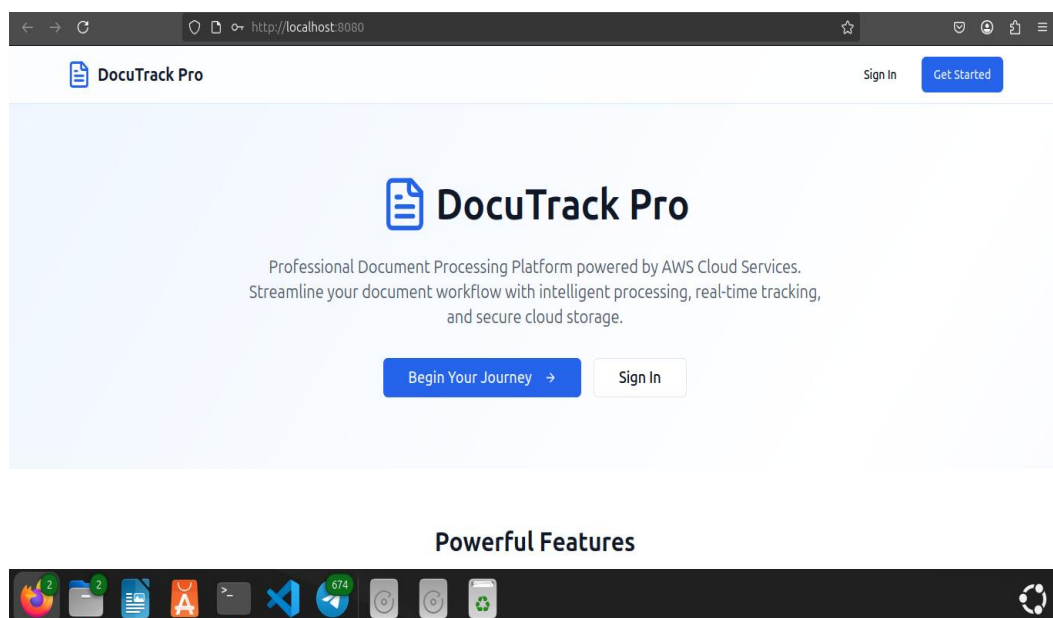
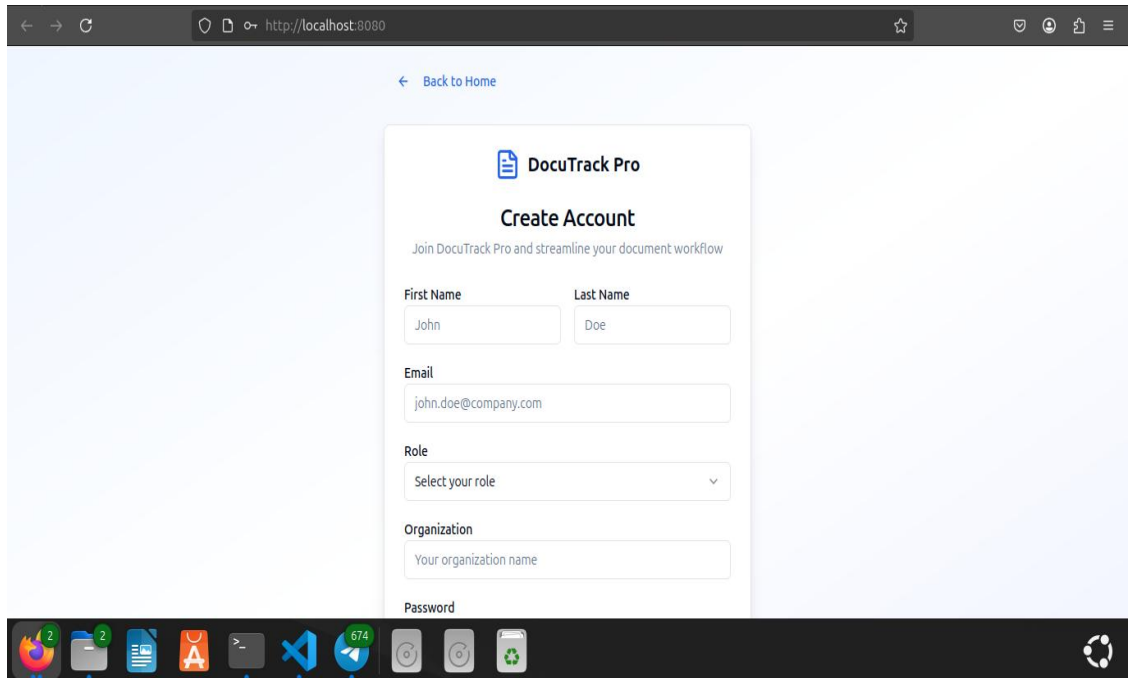


Figure 16:Home Page

❖ Sign Up Page



The screenshot shows a web browser window at `http://localhost:8080` displaying the 'Sign Up Page' for 'DocuTrack Pro'. The page has a light blue background with a white central form. At the top of the form is the 'DocuTrack Pro' logo and the title 'Create Account'. Below the title is a subtitle: 'Join DocuTrack Pro and streamline your document workflow'. The form contains several input fields: 'First Name' (with 'John' entered), 'Last Name' (with 'Doe' entered), 'Email' (with 'john.doe@company.com' entered), 'Role' (a dropdown menu with 'Select your role' selected), 'Organization' (with 'Your organization name' entered), and 'Password'. A 'Back to Home' link is located at the top left of the form area. The browser's address bar and a Windows taskbar with various application icons are visible at the bottom of the screenshot.

Figure 17:Sign Up Page

N

❖ Sign In Page

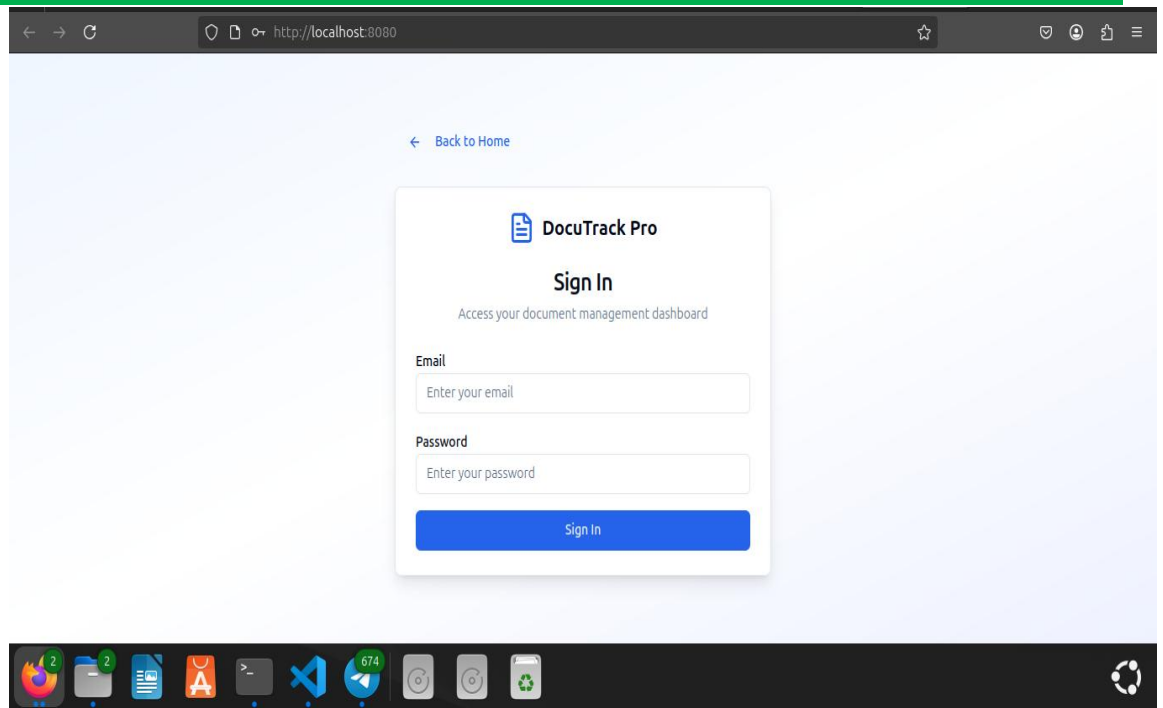


Figure 18:Sign In Page

❖ Deputy Director Dashboard

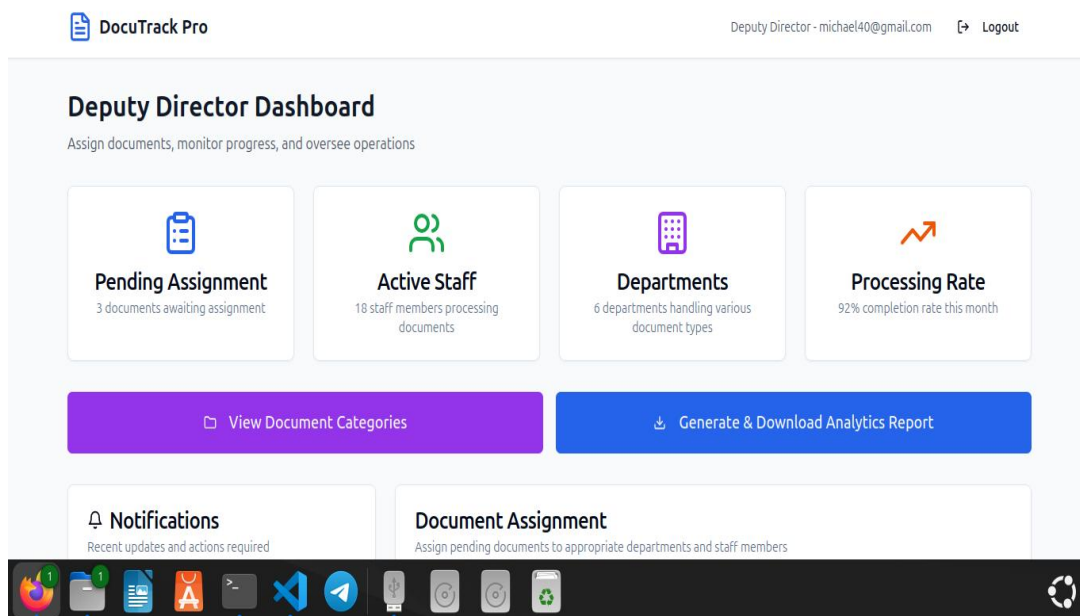


Figure 19:Deputy Director Dashboard

❖ Processing Staff Dashboard

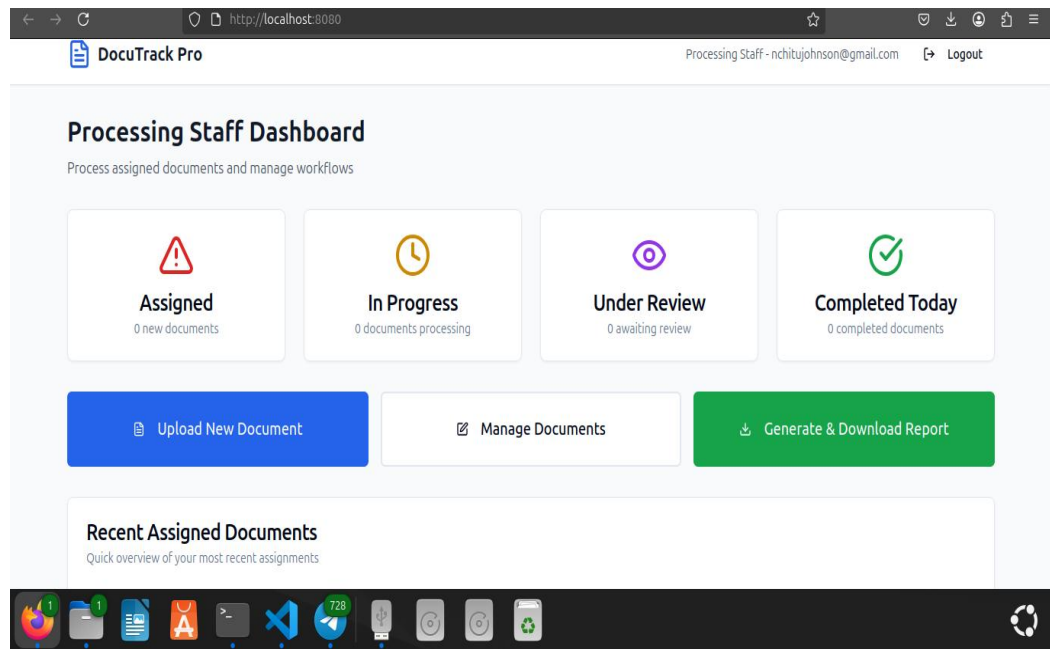


Figure 20: Processing Staff Dashboard

❖ User (Client) Dashboard

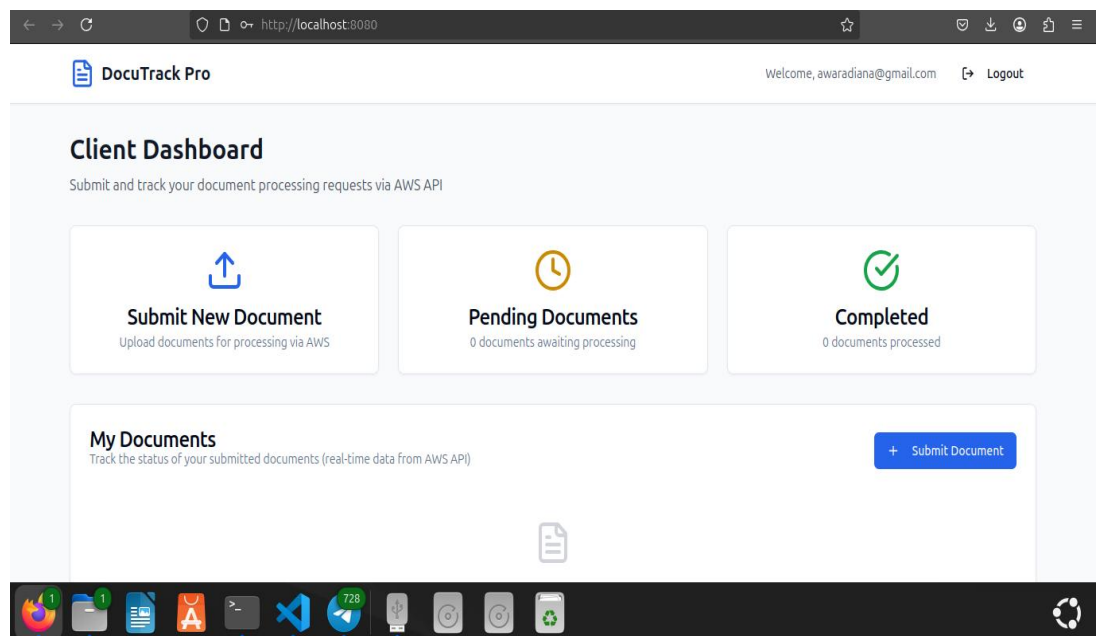


Figure 21: User (Client) Dashboard

3.7. CONCLUSION

In conclusion, the implementation of DocuTrack Pro demonstrates a robust and modern approach to cloud-based document management, leveraging cutting-edge web technologies and the scalability of AWS's serverless architecture. The system was carefully designed with a modular, microservices-based structure, ensuring flexibility, maintainability, and performance. Through a combination of frontend technologies like React, TypeScript, and Tailwind CSS, and backend services including AWS Lambda, API Gateway, and RDS, the platform delivers a seamless and secure experience for users across different roles. Comprehensive database design and clear entity relationships support efficient data handling and traceability. Accessibility, real-time communication, and responsive design were core development priorities, resulting in an intuitive and user-friendly interface. The successful implementation and testing of core functionalities including real-time tracking, secure file management, and role-based dashboards validate the system's effectiveness and readiness for real-world deployment. DocuTrack Pro stands as a scalable, secure, and user-centric solution tailored to modern document processing needs.

GENERAL CONCLUSION

I. INTRODUCTION

This document has presented the full development and deployment workflow of the Smart File Navigator system, a cloud-based document management platform built on AWS. By integrating services such as AWS Lambda, API Gateway, and RDS, the system demonstrates how modern serverless architecture can be used to achieve scalability, security, and reliability. The processes outlined from authentication and role-based access control to secure document storage and retrieval reflect best practices in cloud-native application development.

II. RECALL OF WORK

The DocuTrack Pro project involved developing a comprehensive cloud based document management system designed to streamline document processing workflows for organizations. The system features role-based access control, real-time notifications, comprehensive reporting, and secure file management capabilities.

Key work completed includes:

- ❖ Frontend development using React and TypeScript
- ❖ Backend API development using AWS Lambda and API Gateway
- ❖ Database design and implementation using MySQL
- ❖ Authentication and authorization system implementation
- ❖ Comprehensive testing and deployment procedures

III. ACHIEVEMENTS

The project successfully achieved its core objectives through the complete implementation and thorough testing of all planned features. By leveraging a cloud-native architecture using AWS services, the system ensures high scalability, reliability, and efficient resource management. Enterprise-grade security measures were implemented to maintain data integrity, confidentiality, and compliance with industry standards. A strong emphasis was placed on user experience, resulting in intuitive and user-friendly interfaces tailored to the specific needs of each user role. Additionally, performance optimization efforts led to fast loading times and responsive user interactions, enhancing the overall functionality and satisfaction of the platform.

IV. FURTHER IMPROVEMENT

Potential areas for future improvement of the system include the development of native mobile applications for iOS and Android, allowing users to manage and access documents more conveniently on the go. Incorporating advanced analytics powered by machine learning could provide deeper insights and predictive capabilities to support decision-making and improve workflow efficiency. Enhancing integration capabilities with popular third-party systems such as Office 365 and Google Workspace would increase the platform's versatility and adoption. Further improvements in workflow automation, including intelligent document routing and approval processes, could significantly boost productivity. Adding multi-language support would enable internationalization and make the system accessible to a broader global audience. Lastly, implementing advanced search functionality, including full-text search and AI-powered suggestions, would improve information retrieval and overall user experience.

V. CONCLUSION

In conclusion, the implementation of this system showcases a practical and cost-effective solution for document tracking and management using AWS Free Tier services. It highlights the advantages of serverless infrastructure in reducing operational overhead while maintaining performance. The backend API, developed with Node.js and MySQL, successfully supports key functionalities such as user registration, secure login, and authenticated document operations. Overall, the project proves to be a solid foundation for further development, including extended role management, audit logging, and file versioning in future iterations.

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